

RIDHIMA JAIN

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ECE + Neuroscience student (4.0 GPA) specializing in embedded firmware, sensor fusion, and ML inference on constrained hardware, from autonomous robots to hackathon-winning AI apps.

Education

Wayne State University – Major GPA: 4.0

Detroit, MI | Expected April 2028

B.Eng. Electrical & Computer Engineering | B.S. Neuroscience

Technical Skills

Languages: C, C++, Python, Java, C#

Embedded & Systems: Microcontroller programming (Arduino C), sensor integration, real-time systems, RTOS concepts

Tools: Git, Linux, Docker, CI/CD, modular system design, automated testing

Data / Robotics: Pandas, NumPy, Matplotlib, ROS2

Projects

Autonomous 2D Mapping & Obstacle-Avoidance Robot | *C/C++, Arduino, ROS2*

- Write low-level C/C++ firmware to control hardware components, process sensor data, and generate a real-time 2D occupancy grid
- Architected modular components for sensing, mapping, and path-planning; dynamic obstacle detection recalculates routes in <50ms based on live sensor input

Human Intent Recognition Robot | *Python, C++, Computer Vision, Embedded Systems* [In Progress]

- Developing a robot that infers human intent from behavioral cues (gesture, gaze, motion) using sensor fusion and lightweight ML inference on constrained hardware
- Designing the embedded perception pipeline to balance real-time responsiveness with processing limits, core challenge analogous to TI embedded SDK optimization

Hackathons

NVIDIA Agents for Impact Hackathon — 6th Place | *NVIDIA NeMo Tron, OCR, ACR, React*

2026

- Built an AI productivity app tackling decision fatigue; transcribes and optimizes daily schedules using NVIDIA NeMo Tron-based agents
- Integrated computer vision (OCR/ACR) for real-world input parsing, photographing gym equipment to estimate workout duration, scanning skincare products to recommend usage timing

WayneHacks4 — Sudsly | *React.js, Supabase, Stripe*

2026

- Built a full-stack on-demand laundry pickup and delivery app in 24 hours, with separate customer and driver dashboards and role-based authentication
- Leveraged Supabase for auth, database, and real-time WebSocket subscriptions, syncing live order status between customers and drivers
- Engineered a QR code verification system tied to individual orders, ensuring drivers confirm the correct customer and bag at pickup

Experience

Robotics & Mental Math Instructor – Obotz | Windsor, ON

Aug 2024 – Present

- Taught Arduino C and Python to students aged 8–18, designing hardware-software integration projects mirroring real embedded systems workflows
- Diagnosed and debugged student code live, building systematic troubleshooting skills across C and Python

Junior Code Instructor – Code Ninjas | Windsor, ON

July 2023 – July 2024

- Mentored students in Python, C++, C#, Lua, and Java with emphasis on clean, scalable design patterns and debugging methodology.

Leadership & Activities

Founder, External Relations, Computer Science Association | Robotics Club (ROS2 Team) — collaborative embedded robotics software development | Professional Chair, WWiT – Connecting members with recruiters, speakers, and industry professional
Warrior Bot, AWS Club | Indian Student Association, Fundraising Chair — organized events raising \$20,000 for charity